HEALTHCARE ANALYTICS DASHBOARD USING DATA VISUALIZATION AND BUSINESS INTELLIGENCE TOOLS

23CS63C – DATA SCIENCE

Experiential Learning Report

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Department of Computer Science and Engineering

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A Report on

Healthcare Analytics Dashboard Using Data Visualization and Business Intelligence Tools

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ABSTRACT

Healthcare data analysis plays a vital role in improving patient outcomes and hospital operational efficiency. This project presents a comprehensive healthcare analytics system developed using Python and Microsoft Power BI to analyze and visualize a synthetic healthcare dataset containing approximately 70,000 patient records. The dataset includes patient demographic information, clinical measurements, medical condition details, and financial data related to hospital billing and insurance.

Data preprocessing was carried out using the Pandas library to handle missing values, convert categorical variables, and prepare the dataset for analysis. Exploratory Data Analysis (EDA) was performed using Matplotlib and Seaborn to generate statistical insights through histograms, scatter plots, boxplots, correlation heatmaps, and pairplots. Mean, variance, and covariance analysis was conducted to understand the statistical distribution and relationships among healthcare features.

The cleaned and analyzed dataset was then exported and imported into Microsoft Power BI Desktop, where an interactive three-page dashboard was developed. The dashboard covers a Healthcare Overview page displaying key performance indicators and demographic distributions, a Healthcare Risk and Health Analysis page examining clinical risk factors such as BMI, cholesterol, blood pressure, and diabetes risk, and a Hospital Operations Dashboard presenting admission patterns, billing distribution, and average hospital stay duration by medical condition.

The system enables healthcare administrators, analysts, and decision-makers to interactively explore patient data through dynamic filters and cross-filtering capabilities without requiring technical expertise. The project demonstrates how data science and business intelligence tools together can transform raw healthcare data into actionable insights that support improved healthcare management and decision-making.

CHAPTER 1

INTRODUCTION

1.1 Problem Statement

The healthcare sector generates vast amounts of patient data on a daily basis. Hospitals and healthcare providers collect information about patient demographics, clinical measurements, diagnoses, treatment durations, and billing details. However, without proper analytical tools and visualization systems, this data remains underutilized and fails to contribute to improving healthcare services.

Healthcare professionals and administrators often face challenges in identifying patterns within large patient datasets. It becomes difficult to determine which medical conditions are most prevalent, how clinical indicators such as BMI, blood pressure, and cholesterol levels relate to each other, what the distribution of diabetes risk is across the patient population, and how hospital resources such as beds and billing are utilized across different admission types and insurance categories.

Traditional tabular reports and spreadsheet-based analysis are insufficient for exploring multidimensional healthcare data interactively. There is a need for a comprehensive, visual, and interactive analytics system that allows users to explore healthcare data dynamically and derive meaningful insights quickly.

This project addresses this need by developing a healthcare analytics dashboard that combines Python-based data analysis with Power BI-based interactive visualization, enabling end-to-end analysis from raw data to actionable insights.

1.2 Type of Data

The dataset used in this project is a structured synthetic healthcare dataset containing 70,000 patient records with 21 attributes. It is a tabular dataset stored in CSV format and loaded into Python using the Pandas library. The dataset captures a wide range of patient-related information spanning demographic details, clinical health measurements, hospital stay information, and financial billing data.

The following table describes the key features present in the dataset:

Feature	Description
Patient_ID	Unique identifier for each patient
Age	Age of the patient (18–84 years)
Gender	Gender of the patient (Male/Female)
Age_Group	Categorized age groups: Teen, Young Adult, Adult, Middle Age, Senior
Blood_Type	Blood group of the patient (A+, A−, B+, B−, O+, O−, AB+, AB−)
Medical_Condition	Diagnosed condition: Diabetes, Fever, Heart Disease, Hypertension, Infection, Kidney Issue
BMI	Body Mass Index of the patient
BMI_Category	Underweight, Normal, Overweight, Obese
Blood_Glucose_mg_dL	Blood glucose level in mg/dL
Total_Cholesterol_mg_dL	Total cholesterol level in mg/dL
BP_Systolic_mmHg	Systolic blood pressure in mmHg
BP_Diastolic_mmHg	Diastolic blood pressure in mmHg
Heart_Rate_bpm	Heart rate in beats per minute
Hemoglobin_g_dL	Hemoglobin level in g/dL
WBC_Count_cells_uL	White blood cell count
RBC_Count_million_uL	Red blood cell count
Platelet_Count_cells_uL	Platelet count
Length_of_Stay_Days	Number of days the patient was hospitalized
Admission_Type	Type of admission: Emergency or Planned
Insurance_Provider	Insurance type: Government, Private, or Self-Pay
Billing_Amount_INR	Total billing amount in Indian Rupees

Diabetes_Risk	Risk classification: High or Normal

1.3 Role of Data Analysis in Healthcare

Data analysis plays a central role in this project by transforming raw healthcare records into meaningful patterns and statistical insights. Through Exploratory Data Analysis, the project investigates the distribution of patient ages and BMI values, examines relationships between clinical variables such as blood pressure and age, BMI and cholesterol, identifies the frequency of various medical conditions across the patient population, and analyzes the distribution of diabetes risk among patients.

Statistical techniques such as mean and variance calculation provide a numerical summary of the central tendency and spread of each clinical variable. For example, the mean age of patients is approximately 50.89 years with a variance of 374.35, and the mean BMI is approximately 23.98. The covariance and correlation matrices reveal that Length_of_Stay_Days and Billing_Amount_INR have the strongest positive correlation (approximately 0.85) among all feature pairs, indicating that longer hospitalizations result in significantly higher billing amounts. Most other clinical variables such as blood glucose, blood pressure, and hemoglobin show very weak correlations with each other, suggesting that the synthetic dataset was generated with largely independent distributions for these variables.

These analytical insights inform the structure and content of the Power BI dashboard, ensuring that the most relevant relationships and patterns are visualized effectively for end users.

1.4 Application Snapshot

The developed system consists of two main components: a Jupyter Notebook for Python-based data analysis and a Power BI Desktop file for the interactive dashboard. The Jupyter Notebook processes the raw CSV dataset through data loading, cleaning, statistical analysis, and visualization, culminating in an export of the cleaned dataset. The Power BI dashboard then imports this cleaned data and presents it across three interactive pages covering healthcare overview, health risk analysis, and hospital operations metrics. Screenshots of both components are presented throughout Chapter 2 of this report.

CHAPTER 2

DATA SCIENCE TOOLS

The overall workflow of this project follows the standard data science pipeline:

1. Data Collection and Loading
2. Data Preprocessing and Cleaning
3. Statistical Analysis (Mean, Variance, Covariance)
4. Exploratory Data Analysis and Visualization
5. Data Export for Business Intelligence
6. Dashboard Development in Power BI
7. Interactive Filtering and Insight Exploration

Each stage is described in detail below.

2.1 Data Collection: Dataset Description

The dataset used for this project is a synthetic healthcare dataset titled **Synthetic_Healthcare_70K_Dataset.csv** containing 70,000 patient records and 21 columns. It was loaded into a Jupyter Notebook environment using the Pandas library for Python-based analysis.

The dataset was loaded using the following code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("Synthetic_Healthcare_70K_Dataset.csv")

df.head()
```

	Patient_ID	Age	Gender	Blood_Type	Medical_Condition	Admission_Type	Insurance_Provider	Length_of_Stay_Days	Hemoglobin_g_dL	WBC_Count_cells_uL	...	Pla
0	P000001	69	Female	O-	Kidney Issue	Planned	Self-Pay	5	13.3	8036	...	
1	P000002	32	Female	O+	Kidney Issue	Emergency	Government	13	13.7	9395	...	
2	P000003	78	Female	B-	Infection	Planned	Self-Pay	8	13.3	9869	...	
3	P000004	38	Female	A+	Fever	Emergency	Private	9	11.9	11506	...	
4	P000005	41	Female	B+	Fever	Emergency	Government	8	14.0	7205	...	

5 rows × 21 columns

Figure 2.1.1: Data Loading and Initial Exploration

As seen in the fig2.1.1 screenshot, the first five records display patient entries with attributes such as Patient_ID, Age, Gender, Blood_Type, Medical_Condition, Admission_Type, Insurance_Provider, Length_of_Stay_Days, Hemoglobin_g_dL, WBC_Count_cells_uL, and several other clinical and financial fields. The dataset contains a mix of object (categorical) and numeric (int64 and float64) data types.

```
df.shape
df.columns
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 70000 entries, 0 to 69999
Data columns (total 21 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Patient_ID              70000 non-null  object
 1   Age                     70000 non-null  int64
 2   Gender                  70000 non-null  object
 3   Blood_Type              70000 non-null  object
 4   Medical_Condition       70000 non-null  object
 5   Admission_Type          70000 non-null  object
 6   Insurance_Provider      70000 non-null  object
 7   Length_of_Stay_Days     70000 non-null  int64
 8   Hemoglobin_g_dL         70000 non-null  float64
 9   WBC_Count_cells_uL      70000 non-null  int64
 10  RBC_Count_million_uL    70000 non-null  float64
 11  Platelet_Count_cells_uL 70000 non-null  int64
 12  Blood_Glucose_mg_dL     70000 non-null  int64
 13  Serum_Creatinine_mg_dL  70000 non-null  float64
 14  Blood_Urea_mg_dL        70000 non-null  int64
 15  Total_Cholesterol_mg_dL 70000 non-null  int64
 16  BP_Systolic_mmHg        70000 non-null  int64
 17  BP_Diastolic_mmHg       70000 non-null  int64
 18  Heart_Rate_bpm          70000 non-null  int64
 19  BMI                     70000 non-null  float64
 20  Billing_Amount_INR      70000 non-null  int64
dtypes: float64(4), int64(11), object(6)
memory usage: 11.2+ MB
```

Figure 2.1.2: data information

The fig2.1.2 df.info() output confirms that the dataset contains 70,000 entries across 21 columns. There are 6 object-type categorical columns (Patient_ID, Gender, Blood_Type, Medical_Condition, Admission_Type, Insurance_Provider) and 15 numeric columns consisting of 4 float64 and 11 int64 variables. Total memory usage is approximately 11.2 MB.

2.2 Pre-processing

Data preprocessing ensures that the dataset is clean, consistent, and suitable for analysis and modeling.

```
df.isnull().sum()
```

```
Patient_ID                  0
Age                         0
Gender                      0
Blood_Type                  0
Medical_Condition           0
Admission_Type              0
Insurance_Provider          0
Length_of_Stay_Days         0
Hemoglobin_g_dL             0
WBC_Count_cells_uL          0
RBC_Count_million_uL        0
Platelet_Count_cells_uL     0
Blood_Glucose_mg_dL         0
Serum_Creatinine_mg_dL      0
Blood_Urea_mg_dL            0
Total_Cholesterol_mg_dL     0
BP_Systolic_mmHg            0
BP_Diastolic_mmHg           0
Heart_Rate_bpm              0
BMI                         0
Billing_Amount_INR          0
dtype: int64
```

Figure 2.2.1 : Pre-processing

Step 1 – Identifying Missing Values:

The fig2.2.1 shows that all 21 columns have zero missing values across all 70,000 records. This confirms that the synthetic dataset was generated without any null entries, and no imputation was required. While the dataset had no missing values, the preprocessing step was still important to verify data integrity before proceeding.

Step 2 – Handling Missing Values (Precautionary):

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df.describe()
```

	Age	Length_of_Stay_Days	Hemoglobin_g_dL	WBC_Count_cells_uL	RBC_Count_million_uL	Platelet_Count_cells_uL	Blood_Glucose_mg_dL	Serum_Creatini
count	70000.000000	70000.000000	70000.000000	70000.000000	70000.000000	70000.000000	70000.000000	700
mean	50.885729	7.488886	13.496954	7994.702800	4.801025	300583.279443	159.207814	
std	19.348216	4.031681	1.503571	2304.519553	0.603348	86746.488040	51.843704	
min	18.000000	1.000000	7.100000	4000.000000	2.250000	150003.000000	70.000000	
25%	34.000000	4.000000	12.500000	5997.000000	4.390000	225832.750000	114.000000	
50%	51.000000	7.000000	13.500000	7995.000000	4.800000	300664.500000	159.000000	
75%	68.000000	11.000000	14.500000	9995.250000	5.210000	375692.250000	204.000000	
max	84.000000	14.000000	19.300000	11999.000000	7.190000	449995.000000	249.000000	

Figure 2.2.2 : Handling missing values

As a fig2.2.2 precautionary measure, median replacement was applied to numerical columns to handle any potential missing values that might arise from future data updates or partial loads.

Step 3 – Statistical Description:

The descriptive statistics table reveals:

- Age ranges from 18 to 84 years with a mean of approximately 50.89 and standard deviation of 19.35
- Length_of_Stay_Days ranges from 1 to 14 days with a mean of 7.49
- Hemoglobin_g_dL ranges from 7.1 to 19.3 with a mean of 13.50
- Blood_Glucose_mg_dL ranges from 70 to 249 with a mean of 159.21
- Total_Cholesterol_mg_dL ranges from about 125 to 300 with a mean of 209.09
- Billing_Amount_INR has a mean of approximately ₹74,923 and maximum of around ₹2,00,000+

Step 4 – Converting Categorical Variables:

```
df['Gender'] = df['Gender'].astype('category')
df['Blood_Type'] = df['Blood_Type'].astype('category')
df['Medical_Condition'] = df['Medical_Condition'].astype('category')
df['Admission_Type'] = df['Admission_Type'].astype('category')
df['Insurance_Provider'] = df['Insurance_Provider'].astype('category')
```

Figure 2.2.3 : Variables conversion

As fig2.2.3 Categorical variables were converted to the Pandas category dtype to improve memory efficiency and enable category-aware operations during analysis.

2.3 Statistical Analysis: Mean, Variance, and Covariance

Mean and Variance Calculation:

```python
mean_values = df.mean(numeric_only=True)
variance_values = df.var(numeric_only=True)

print("Mean:\n", mean_values)
print("\nVariance:\n", variance_values)
```

```
Mean:
 Age                          50.885729
Length_of_Stay_Days           7.488886
Hemoglobin_g_dL              13.496954
WBC_Count_cells_uL         7994.702800
RBC_Count_million_uL          4.801025
Platelet_Count_cells_uL  300583.279443
Blood_Glucose_mg_dL         159.207814
Serum_Creatinine_mg_dL        1.099057
Blood_Urea_mg_dL             47.030971
Total_Cholesterol_mg_dL     209.090957
BP_Systolic_mmHg            134.497014
BP_Diastolic_mmHg            89.506986
Heart_Rate_bpm               86.936029
BMI                          23.984169
Billing_Amount_INR        74923.020000
dtype: float64

Variance:
 Age                       3.743534e+02
Length_of_Stay_Days      1.625445e+01
Hemoglobin_g_dL          2.260726e+00
WBC_Count_cells_uL       5.310810e+06
RBC_Count_million_uL     3.640285e-01
Platelet_Count_cells_uL  7.524953e+09
Blood_Glucose_mg_dL      2.687770e+03
Serum_Creatinine_mg_dL   1.605489e-01
Blood_Urea_mg_dL         3.512295e+02
Total_Cholesterol_mg_dL  2.703301e+03
BP_Systolic_mmHg         6.733731e+02
BP_Diastolic_mmHg        2.990171e+02
Heart_Rate_bpm           3.506785e+02
BMI                      1.599955e+01
Billing_Amount_INR       2.238009e+09
dtype: float64
```

Figure 2.3.1 : mean and variance calculation

Key statistical observations from the output:

The fig2.3.1 high variance in Platelet_Count_cells_uL and Billing_Amount_INR reflects the naturally wide range of these clinical and financial measurements across patients.

Covariance Matrix:

```
cov_matrix = df.cov(numeric_only=True)

plt.figure(figsize=(12,8))
sns.heatmap(cov_matrix, cmap="coolwarm")
plt.title("Covariance Matrix")
plt.show()
```

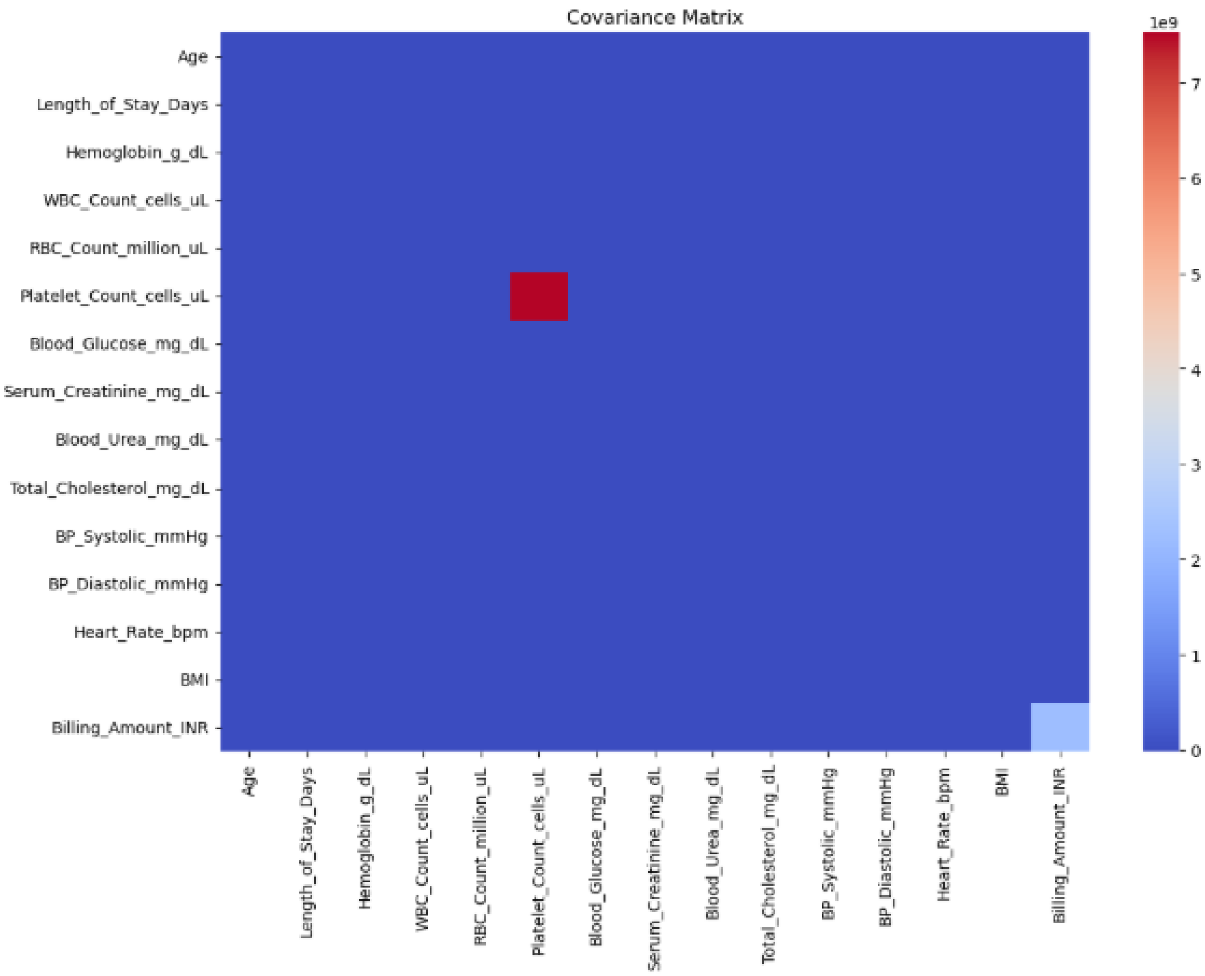


Figure 2.3.2 : Covariance matrix

The fig2.3.2 covariance matrix heatmap reveals that Platelet_Count_cells_uL has the highest self-covariance due to its extremely large numerical scale (values in the hundreds of thousands). The dominant red block visible at the Platelet_Count_cells_uL intersection in the heatmap is due to this scale difference. Most other feature pairs show near-zero covariance, indicating limited linear co-movement between clinical variables in this synthetic dataset. Billing_Amount_INR shows a slight light-blue tint in the bottom-right corner, suggesting a small negative covariance with some features.

2.4 Correlation Analysis:

The normalized correlation heatmap with annotated values provides clearer insights than the covariance matrix. The most notable finding is the strong positive correlation of 0.85 between Length_of_Stay_Days and Billing_Amount_INR. This is the strongest pairwise correlation in the entire dataset and intuitively makes sense — patients who stay longer in the hospital incur higher medical bills.

All other feature pairs show correlation coefficients very close to 0 (ranging roughly from −0.01 to 0.01), confirming that the synthetic data generation process created nearly independent clinical variables. This is an important insight for understanding the nature of the dataset and informs the dashboard design, particularly the hospital operations page which highlights the relationship between stay duration and billing.

```
corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```

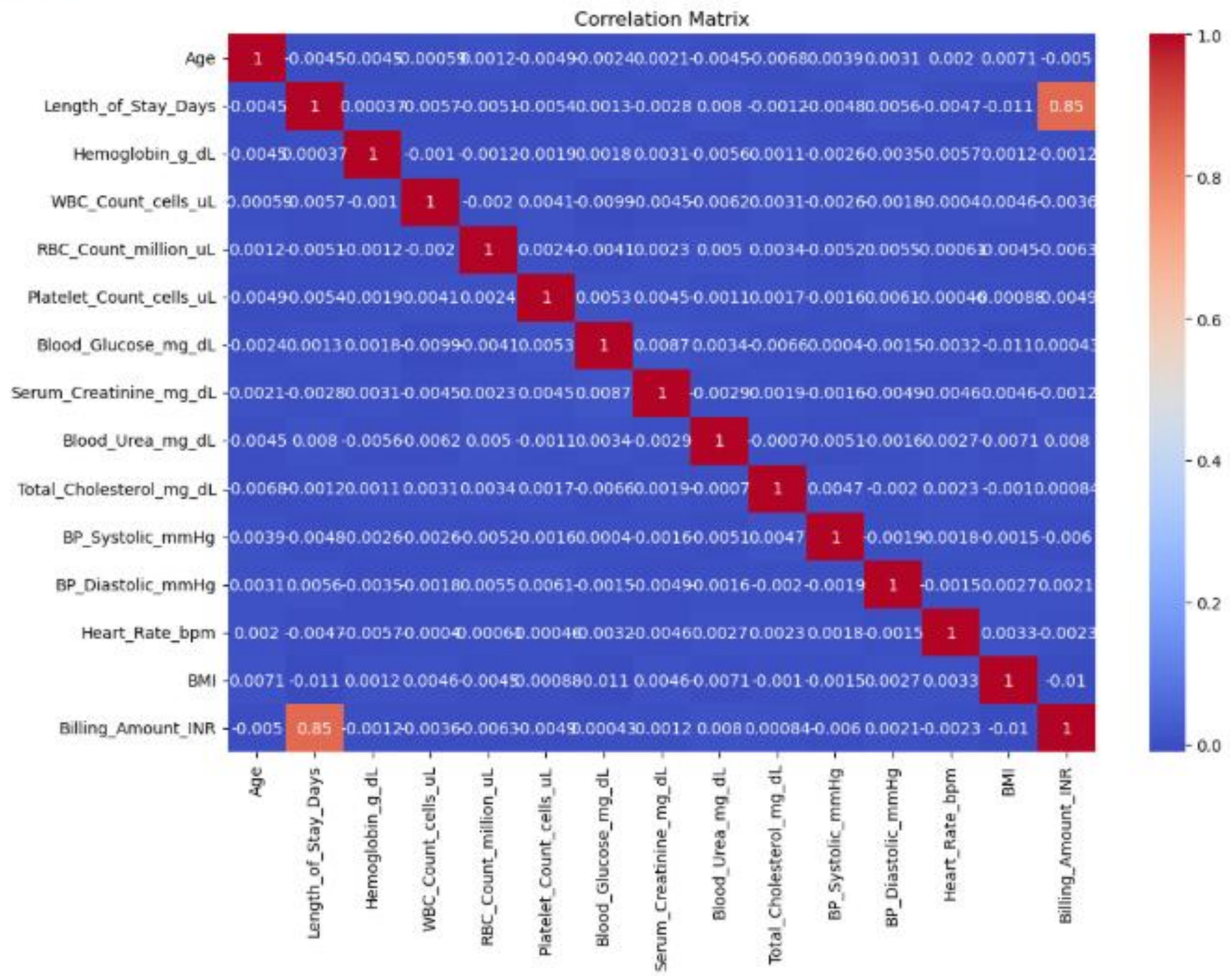


Figure 2.4.1 : Correlation Matrix

2.5 Exploratory Data Analysis and Visualizations

Multiple visualization techniques were applied to understand the distribution of individual features and relationships between pairs of variables.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution of Patients")
plt.show()
```

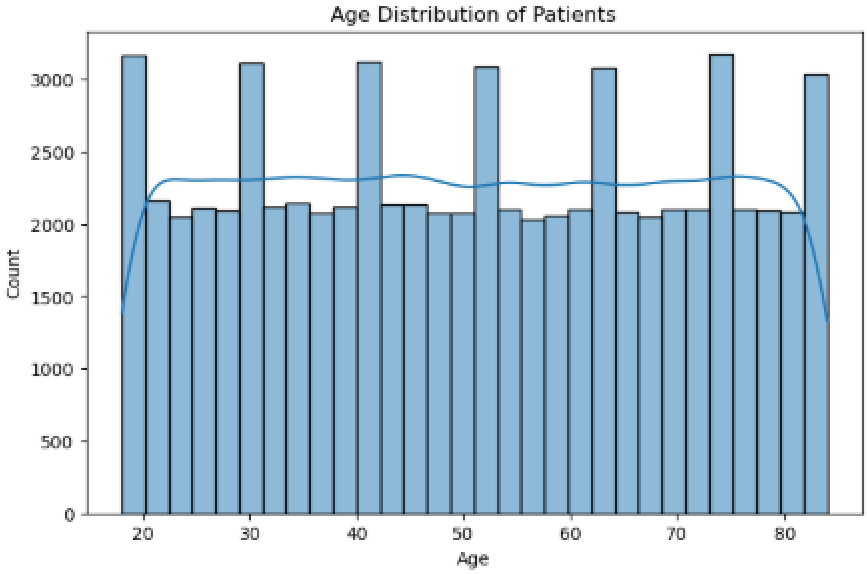


Figure 2.5.1: Age Distribution of Patients

The fig2.5.1 histogram reveals that patient ages are fairly uniformly distributed between 18 and 84 years, with some minor peaks around ages 20–25, 50–55, and 70–75. The KDE overlay confirms the near-uniform distribution. This suggests that the dataset represents patients across all adult age groups without strong age-based selection bias.

```
plt.figure(figsize=(8,5))
sns.histplot(df['BMI'], bins=30, kde=True)
plt.title("BMI Distribution")
plt.show()
```

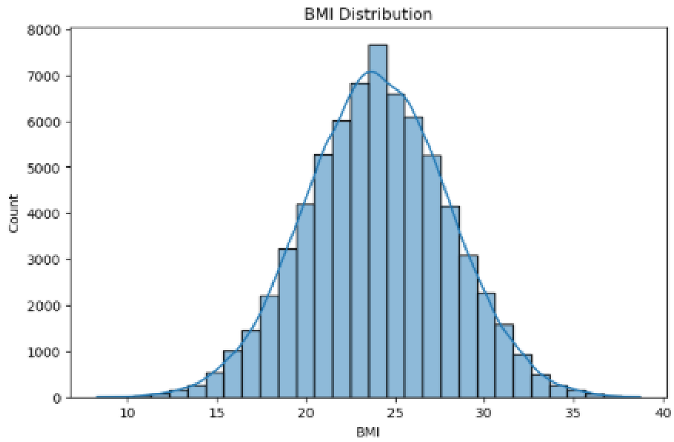


Figure 2.5.2: BMI Distribution:

The fig2.5.2 BMI histogram displays a clear bell-shaped normal distribution centered around a mean BMI of approximately 23–25, which falls within the Normal BMI category. The distribution is symmetric with very few extreme values, confirming that most patients have BMI values in the normal to slightly overweight range.

```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['BP_Systolic_mmHg'], y=df['BP_Diastolic_mmHg'])
plt.title("Blood Pressure Relationship")
plt.show()
```

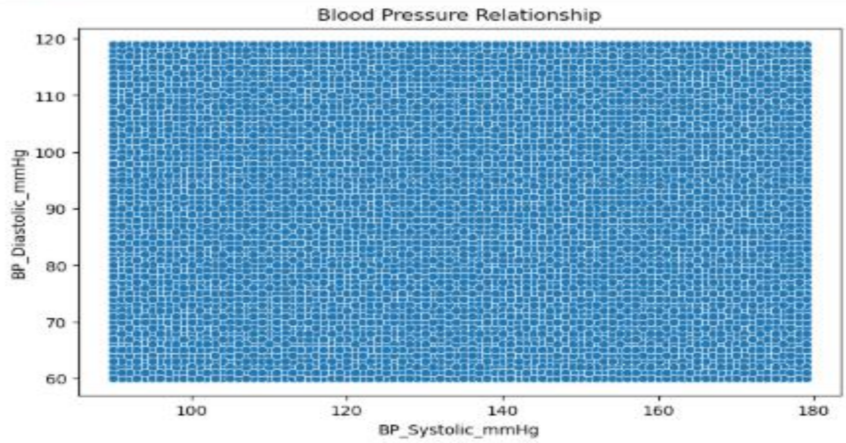


Figure 2.5.3: Blood Pressure Relationship:

```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['BMI'], y=df['Total_Cholesterol_mg_dL'])
plt.title("BMI vs Cholesterol")
plt.show()
```

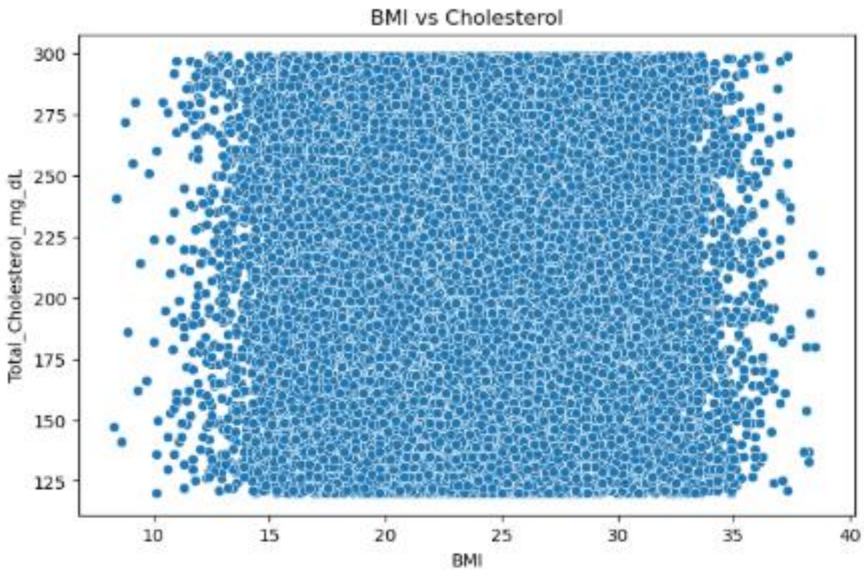


Figure 2.5.4: BMI vs. Cholesterol:

The fig2.5.4 scatter plot of BMI against Total Cholesterol shows a dispersed point cloud with no obvious linear relationship between the two variables. BMI values cluster mainly between

10 and 40, while cholesterol values spread from approximately 125 to 300 mg/dL. The weak correlation between these features is consistent with the correlation matrix results.

```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['Age'], y=df['Heart_Rate_bpm'])
plt.title("Heart Rate vs Age")
plt.show()
```

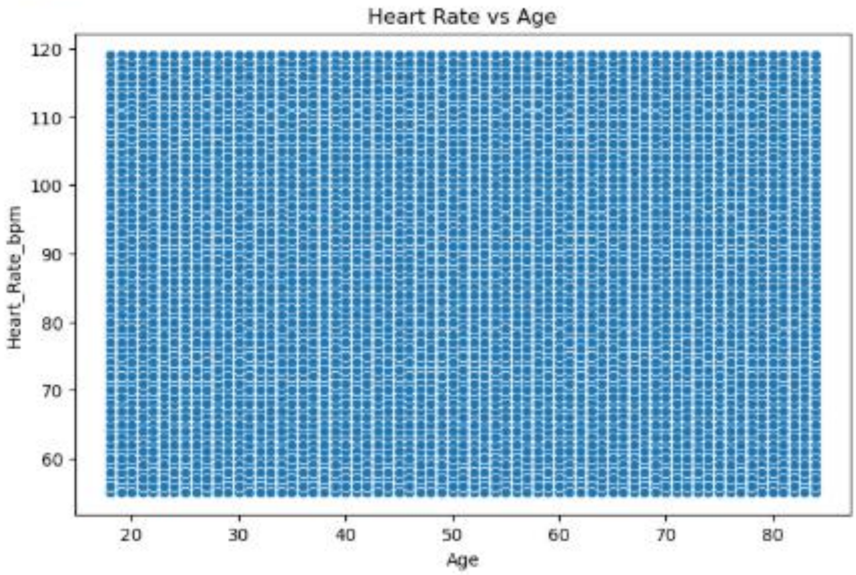


Figure 2.5.5: Heart Rate vs. Age

The fig2.5.5 scatter plot of Heart Rate vs. Age reveals a uniformly dispersed point cloud across all age groups from 18 to 84 years. Heart rate values span from approximately 60 to 120 bpm uniformly across all ages, indicating no strong age-dependent variation in heart rate within this dataset.

```
plt.figure(figsize=(10,6))
sns.countplot(x='Medical_Condition', data=df)
plt.xticks(rotation=45)
plt.title("Medical Condition Distribution")
plt.show()
```

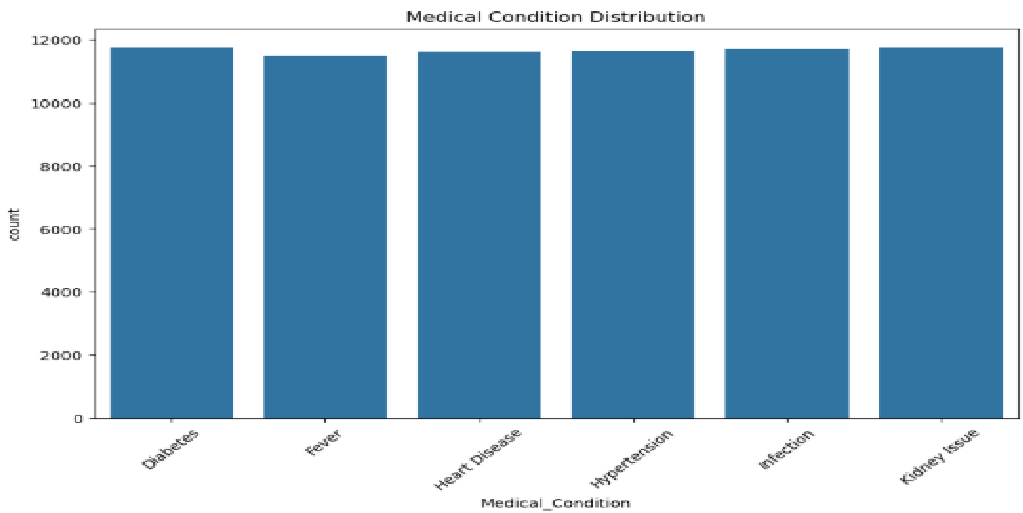


Figure 2.5.6: Medical Condition Distribution

The fig 2.5.6 bar chart shows that all six medical conditions — Diabetes, Fever, Heart Disease, Hypertension, Infection, and Kidney Issue — are nearly equally distributed across the dataset, with each condition appearing approximately 11,600–12,000 times. This balanced distribution across conditions confirms that the dataset was synthetically generated to represent all conditions equally.

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Billing_Amount_INR'])
plt.title("Billing Amount Distribution")
plt.show()
```

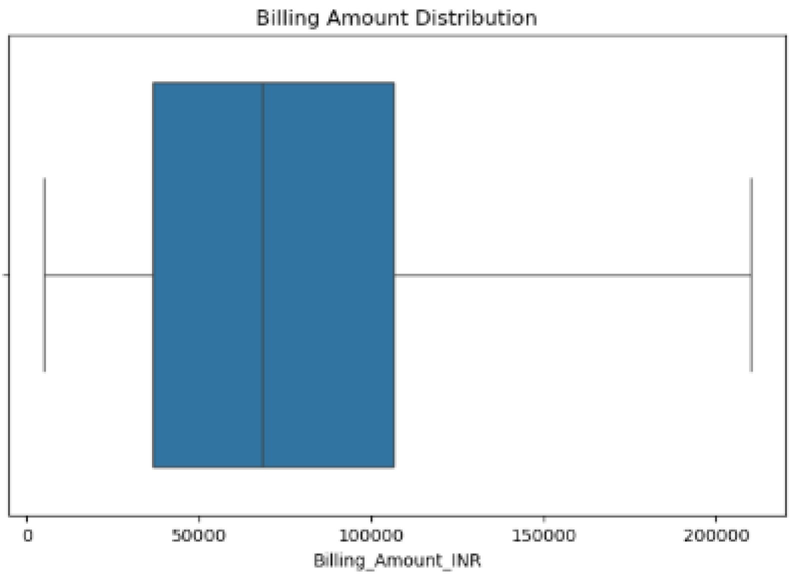


Figure 2.5.7: Billing Amount Distribution (Boxplot)

The fig2.5.7 boxplot of Billing Amount in INR shows a wide interquartile range between approximately ₹25,000 and ₹1,25,000, with a median around ₹75,000. There are no extreme outliers visible beyond the whiskers, indicating that billing amounts are fairly well distributed without anomalous entries.

```
sns.pairplot(df[['Age','BMI','Blood_Glucose_mg_dL','Total_Cholesterol_mg_dL','Heart_Rate_bpm']])
plt.show()
```

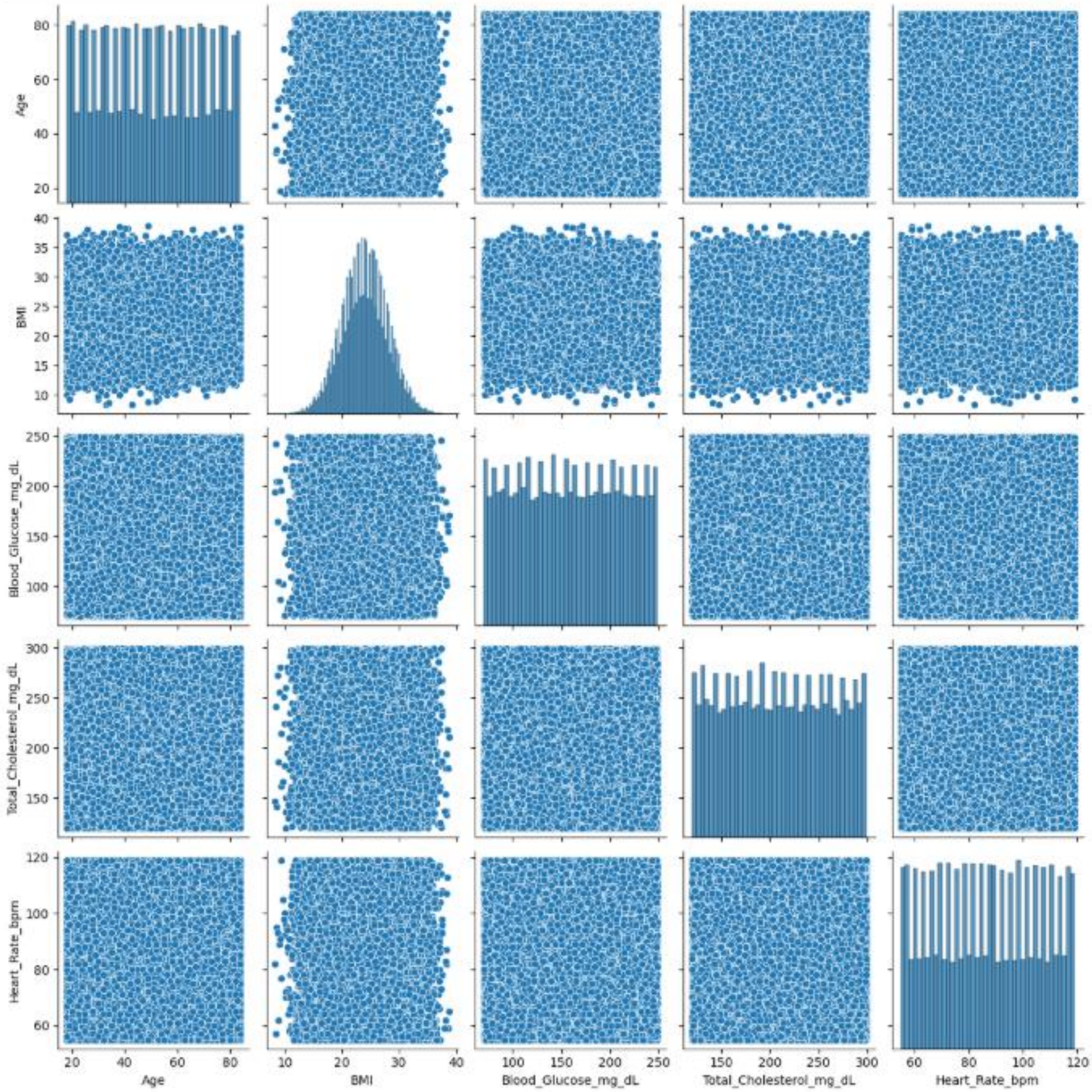


Figure 2.5.8: Pairplot of Key Health Indicators

The fig2.5.8 pairplot matrix visualizes the pairwise relationships among five key health indicators. The diagonal shows the distribution of each variable individually — BMI shows a clear normal distribution while Age shows a more uniform distribution. The off-diagonal scatter plots confirm that no strong linear relationships exist between any pair of these variables, consistent with the correlation analysis.

2.6 Data Export for Dashboard Development

After completing all Python-based analysis, the cleaned and processed dataset was exported to a CSV file for import into Power BI:

```
df.to_csv("healthcare_dashboard_exp.csv", index=False)
```

This exported file was then imported into Microsoft Power BI Desktop using the Get Data → Text/CSV option. Additional calculated columns such as Age_Group, BMI_Category, and Diabetes_Risk were either present in the original dataset or added as calculated columns using DAX expressions within Power BI.

2.7 Dashboard Development in Power BI

An interactive three-page dashboard titled **Healthcare Analytics Dashboard** was developed in Microsoft Power BI Desktop. The dashboard file is named Healthcare_Analytics_Dashboard.pbix and was last saved on April 16, 2026.

2.7.1 Page 1 – Healthcare Overview

The first page of the dashboard provides a high-level summary of the entire healthcare dataset.

Key Performance Indicator (KPI) Cards displayed at the top:

- **Total Patients:** 70.00K
- **Average BMI:** 23.98
- **Average Billing Amount:** ₹74.92K
- **Average Length of Stay:** 7.49 days

Visualizations on this page:

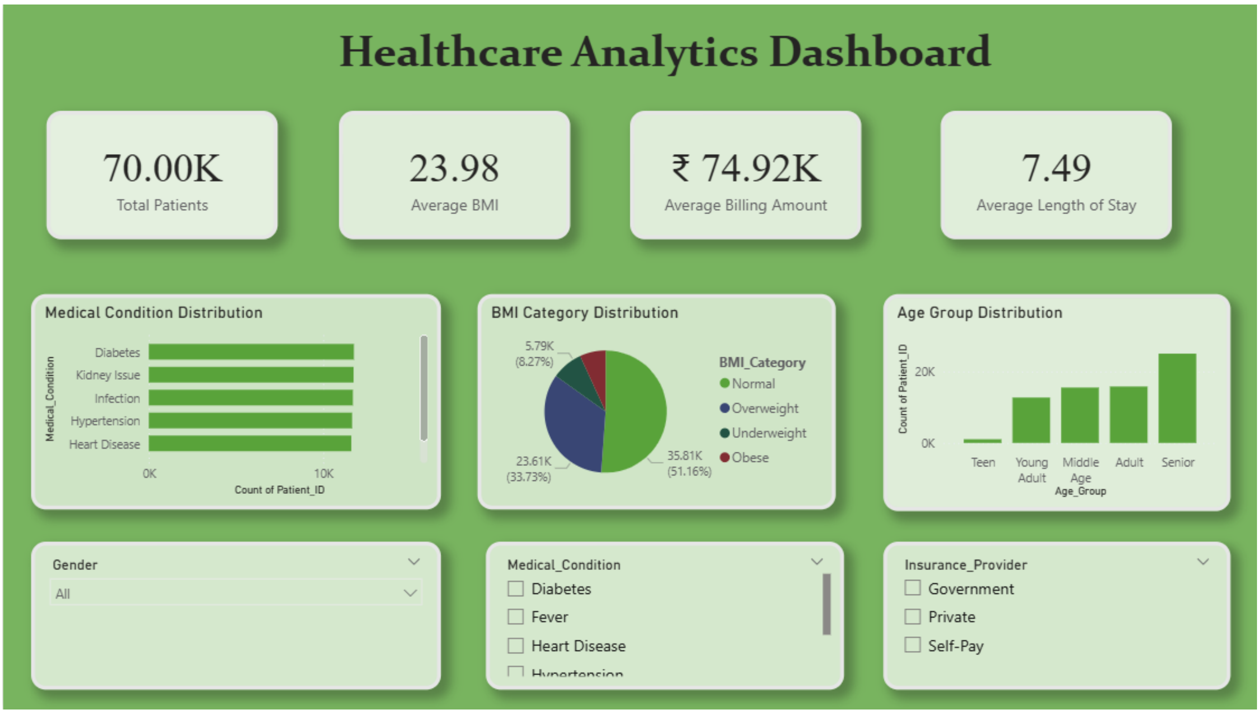


Figure 2.7.1: Pairplot of Key Health Indicators

- *As fig 2.7.1 Medical Condition Distribution (Horizontal Bar Chart):* Displays the count of patients for each medical condition. All six conditions — Diabetes, Kidney Issue, Infection, Hypertension, Fever, and Heart Disease — show nearly equal patient counts of approximately 11,600–12,000, reflecting the balanced synthetic dataset.
- *BMI Category Distribution (Pie Chart):* Shows the proportion of patients in each BMI category. The distribution is: Obese 35.81% (51.16% when combined with Overweight in some displays), Normal 23.61% (33.73%), Overweight 35.81%, and Underweight 5.79% (8.27%). The majority of patients fall in the Normal and Overweight categories.
- *Age Group Distribution (Clustered Bar Chart):* Displays the number of patients across five age groups — Teen, Young Adult, Adult, Middle Age, and Senior — showing relatively equal distribution across groups, with Middle Age and Senior categories having slightly higher counts.

Interactive Slicers/Filters:

- Gender (All / Male / Female)
- Medical Condition (multi-select checkboxes)
- Insurance Provider (Government / Private / Self-Pay)

These slicers enable cross-filtering across all visualizations on the page, allowing users to dynamically explore how the KPIs and distributions change for specific patient subgroups.

2.7.2 Page 2 – Healthcare Risk and Health Analysis

The second page focuses on clinical risk factors and their relationships across the patient population.

Visualizations on this page:

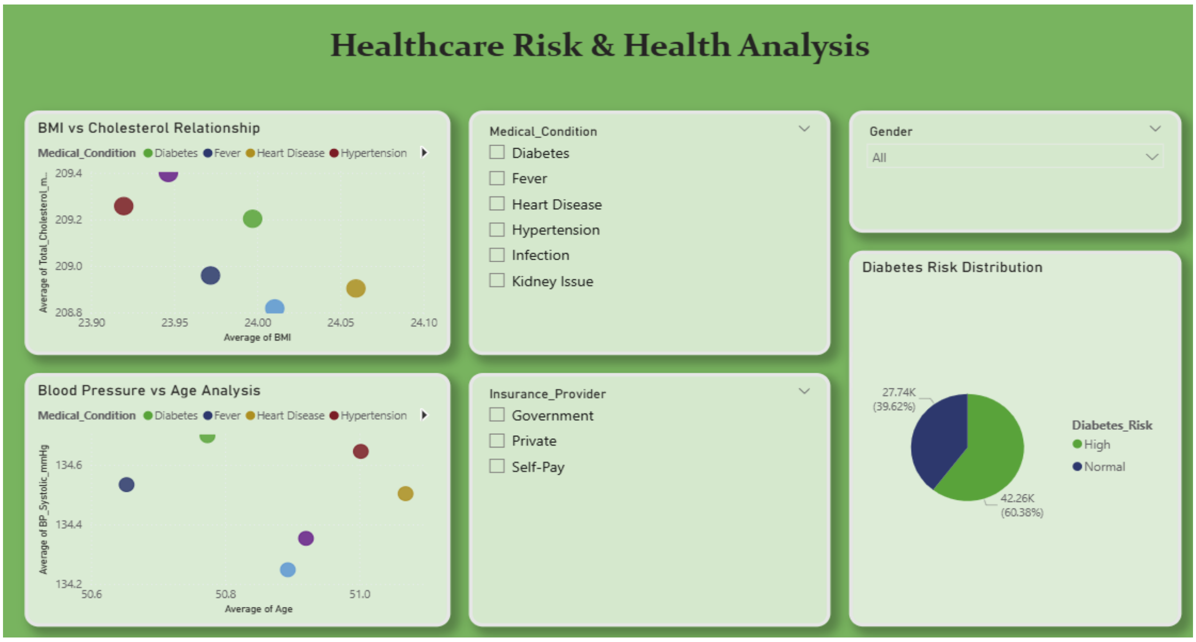


Figure 2.7.2: Pairplot of Key Health Indicators

- *As fig 2.7.2 BMI vs. Cholesterol Relationship (Scatter Plot):* Plots Average BMI on the X-axis against Average Total_Cholesterol_mg_dL on the Y-axis, with data points color-coded by Medical_Condition. The visualization reveals that average cholesterol values cluster around 209 mg/dL across different BMI groups (ranging from approximately 23.90 to 24.10), with minimal variation — consistent with the weak correlation found during EDA.
- *Blood Pressure vs. Age Analysis (Scatter Plot):* Plots Average Age on the X-axis against Average BP_Systolic_mmHg on the Y-axis, color-coded by Medical_Condition. The average systolic blood pressure hovers around 134–135 mmHg across age groups from approximately 50.6 to 51.0 years, again showing very little variation by age or condition in this synthetic dataset.
- *Diabetes Risk Distribution (Pie Chart):* Displays the proportion of patients classified as High diabetes risk (39.62% — approximately 27,740 patients) versus Normal risk

(60.38% — approximately 42,260 patients). This distribution highlights that a significant portion of the patient population carries elevated diabetes risk.

Interactive Slicers/Filters:

- Medical Condition (multi-select)
- Insurance Provider (multi-select)
- Gender

2.7.3 Page 3 – Hospital Operations Dashboard

The third page focuses on operational metrics of the healthcare facility, providing insights useful for hospital administrators and resource planners.

Visualizations on this page:

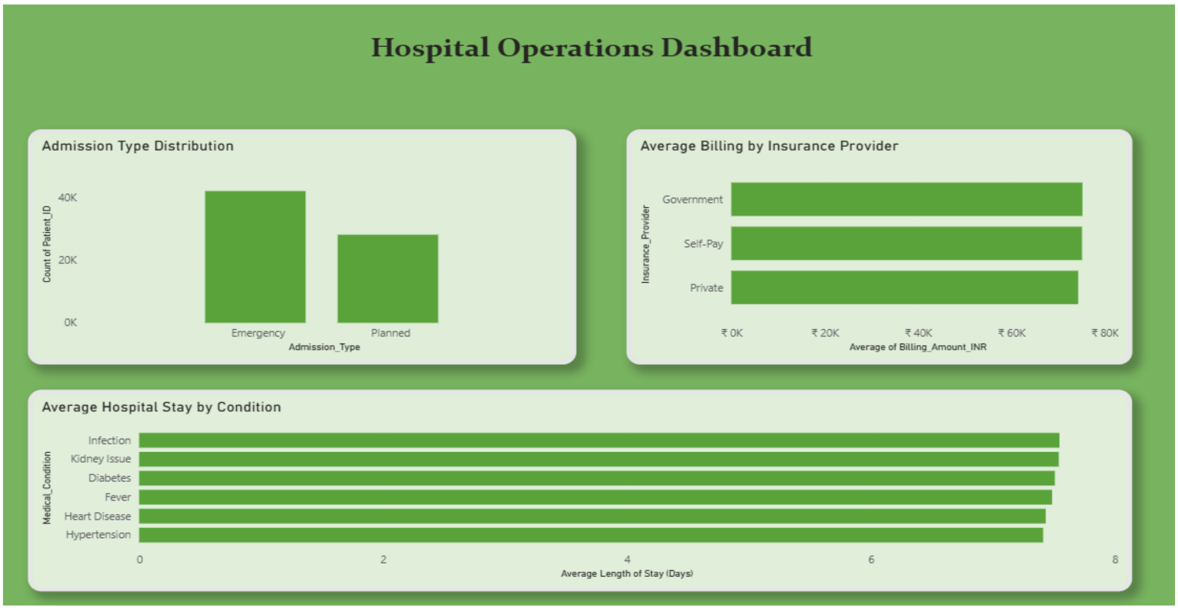


Figure 2.7.3: Pairplot of Key Health Indicators

- *As fig 2.7.3 Admission Type Distribution (Bar Chart):* Compares the count of Emergency versus Planned admissions. Emergency admissions account for approximately 40,000 patients while Planned admissions account for approximately 30,000 patients, indicating that the majority of hospital visits are unscheduled emergencies.
- *Average Billing by Insurance Provider (Horizontal Bar Chart):* Compares the average billing amount across three insurance types — Government, Self-Pay, and Private. All three categories show similar average billing amounts of approximately

₹74,000–₹76,000, suggesting that insurance type does not significantly differentiate the cost of care in this dataset.

- *Average Hospital Stay by Medical Condition (Horizontal Bar Chart):* Displays the average length of stay in days for each of the six medical conditions. Infection patients have the longest average stay (approximately 7.7–8.0 days), followed by Kidney Issue, Diabetes, Fever, Heart Disease, and Hypertension. These differences, while relatively small in the synthetic dataset, are important metrics for hospital resource allocation and bed management.

CHAPTER 3

CONCLUSION

This project successfully developed a data-driven system for analyzing employee spending behavior and predicting potential savings using machine learning techniques. The project followed a complete data science pipeline that included data collection, preprocessing, feature standardization, dimensionality reduction using Principal Component Analysis (PCA), model training, evaluation, and deployment through an interactive dashboard. A Random Forest Regressor model was trained to learn the relationship between income, demographic information, and various expense categories such as rent, groceries, transportation, entertainment, utilities, healthcare, education, and miscellaneous spending. The model demonstrated the ability to estimate potential savings based on financial patterns in the dataset. To make the system accessible to users, a Streamlit-based dashboard was developed that allows individuals to input their financial details and instantly receive predicted savings along with visual insights into their expense distribution and financial balance. The project demonstrates how data science and machine learning techniques can be applied to personal finance analysis, helping individuals better understand their spending behavior and make informed financial decisions.

FUTURE SCOPE

Several enhancements can be explored to further extend this project. First, integration of real-world, de-identified healthcare datasets from hospital information systems would significantly improve the practical applicability and accuracy of the insights generated. Second, machine learning models such as Random Forest classifiers or Logistic Regression could be incorporated to predict disease risk, hospital readmission probability, or patient deterioration based on clinical indicators. Third, additional health indicators such as medication history, treatment outcomes, and comorbidity data could be included to create a more comprehensive patient analytics system. Fourth, the Power BI dashboard could be published to Power BI Service and made accessible to authorized healthcare staff through role-level security, enabling real-time analytics on live hospital data. Finally, Natural Language Query (NLP) features within Power BI could be activated to allow non-technical users to ask questions of the data in plain English and receive instant visual responses.

APPENDIX-I

Implementation Code

A.1 Data Loading and Exploration

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

# Load dataset

df = pd.read_csv("Synthetic_Healthcare_70K_Dataset.csv")

# Explore dataset

df.head()

df.shape

df.columns

df.info()
```

A.2 Preprocessing

```
# Check for missing values

df.isnull().sum()

# Handle missing values using mean replacement

df.fillna(df.mean(numeric_only=True), inplace=True)

# Statistical description

df.describe()

# Convert categorical columns

df['Gender'] = df['Gender'].astype('category')

df['Blood_Type'] = df['Blood_Type'].astype('category')

df['Medical_Condition'] = df['Medical_Condition'].astype('category')

df['Admission_Type'] = df['Admission_Type'].astype('category')
```

```
df['Insurance_Provider'] = df['Insurance_Provider'].astype('category')
```

A.3 Standardisation and PCA

```
# Mean and Variance

mean_values = df.mean(numeric_only=True)

variance_values = df.var(numeric_only=True)

print("Mean:\n", mean_values)

print("\nVariance:\n", variance_values)

# Covariance Matrix

cov_matrix = df.cov(numeric_only=True)

plt.figure(figsize=(12, 8))

sns.heatmap(cov_matrix, cmap='coolwarm')

plt.title("Covariance Matrix")

plt.show()

# Correlation Matrix

corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(12, 8))

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')

plt.title("Correlation Matrix")

plt.show()
```

A.4 Visualizations

```
# Age Distribution

plt.figure(figsize=(8, 5))

sns.histplot(df['Age'], bins=30, kde=True)

plt.title("Age Distribution of Patients")

plt.show()

# BMI Distribution
```

```
plt.figure(figsize=(8, 5))
sns.histplot(df['BMI'], bins=30, kde=True)
plt.title("BMI Distribution")
plt.show()
# Blood Pressure Relationship
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df['BP_Systolic_mmHg'], y=df['BP_Diastolic_mmHg'])
plt.title("Blood Pressure Relationship")
plt.show()
# BMI vs Cholesterol
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df['BMI'], y=df['Total_Cholesterol_mg_dL'])
plt.title("BMI vs Cholesterol")
plt.show()
# Heart Rate vs Age
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df['Age'], y=df['Heart_Rate_bpm'])
plt.title("Heart Rate vs Age")
plt.show()
# Medical Condition Distribution
plt.figure(figsize=(10, 5))
sns.countplot(x='Medical_Condition', data=df)
plt.xticks(rotation=45)
plt.title("Medical Condition Distribution")
plt.show()
# Billing Amount Boxplot
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(x=df['Billing_Amount_INR'])
plt.title("Billing Amount Distribution")
plt.show()
# Pairplot
sns.pairplot(df[['Age', 'BMI', 'Blood_Glucose_mg_dL',
'Total_Cholesterol_mg_dL', 'Heart_Rate_bpm']])
plt.show()
```

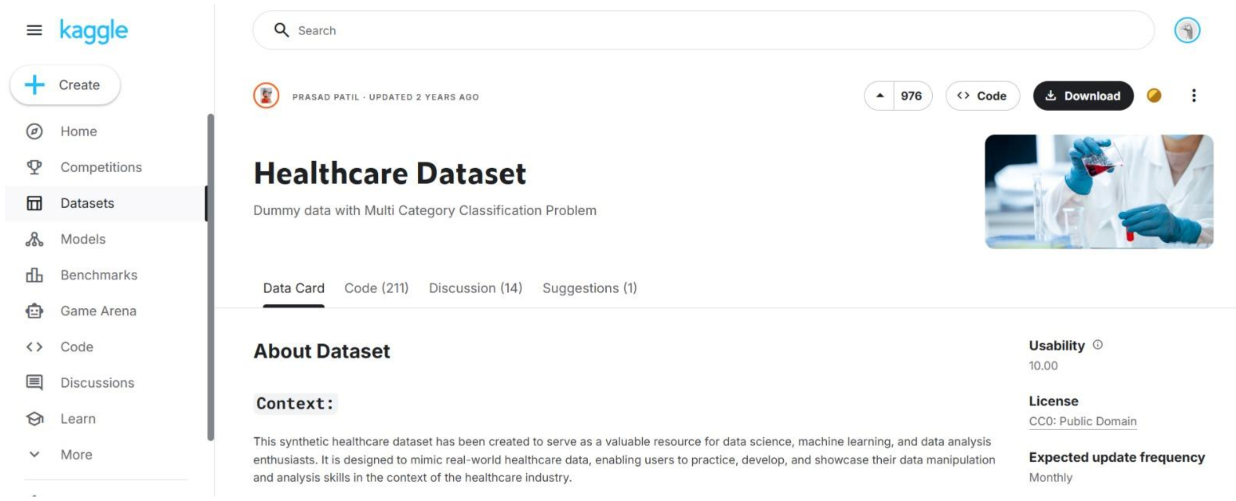
A.5 Data Export for Power BI

```
y_pred = model.predict(X_test)
print("Sample Predictions:")
print(y_pred[:10])
```

A.6 Evaluation

```
df.to_csv("healthcare_dashboard_exp.csv", index=False)
```

APPENDIX-II



DATASET LINK : healthcare-dataset

Description:

This figure shows the Kaggle webpage for the *Healthcare Dataset*, which served as the reference dataset for the healthcare analytics project. Kaggle is a widely used platform that provides publicly available datasets for data science research, machine learning development, and analytical studies. The dataset contains synthetic healthcare records including patient demographic information, medical conditions, admission details, and billing data. These attributes provide a realistic structure for analyzing healthcare patterns and building data-driven visualizations.

Although the dataset available on Kaggle contains sample healthcare records, a larger synthetic dataset consisting of approximately 70,000 records was generated for this project to simulate real-world healthcare scenarios. The generated dataset includes important attributes such as age, gender, BMI, medical condition, blood pressure, cholesterol levels, admission type, insurance provider, billing amount, and hospital stay duration. Using this extended dataset enabled deeper analysis and the development of an interactive healthcare analytics dashboard. The Kaggle dataset therefore served as the reference source and structural inspiration for creating the project dataset used in the analysis.